Climate and environmental attention: a news-based composite indicator

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Outline of the talk

Part 1 The description of a novel methodology to evaluate the *relative semantic relevance*, not the favorability, of a keyword.

Part 2 Definition of an index to measure *Climate and environmental attention* by aggregating the relative semantic importance of a selection of topical keywords.

Part 3 Assessing the impact of the above index in financial markets.

The talk is based on ongoing work with Andrea Fronzetti Colladon, Barbara Guardabascio, Marco Patacca, and Ludovica Segneri.

Textual analysis and applications in Finance

Textual analysis allows the extraction, through automated algorithms, of valuable information and meaningful insights from large volumes of text data, contributing so to better decision making.

It plays a significant role in sentiment analysis and topic modeling in which researchers aim to gauge public interest and opinion from different sources, and it has been widely applied in empirical finance to measure investor sentiment or attention:

- Assessing the impact on financial returns and volatility to define and predict market trends.
- Helping making trading decisions based on news and event data and related sentiment.
- Extract relevant information from loan applications or credit reports to measure credit risk.

Sentiment analysis and topic modeling

The definition of sentiment and attention within the financial literature is not unique and has evolved over time.

- observable market proxies for sentiment such as NYSE share turnover, dividend premium, and Call-Put Ratio (Baker and Wurgler, 2006)
- specific measures of investor sentiment extracted from specialized media contents (Tetlock, 2007)
  - sentiment as an attention measure (Da et al., 2011) (volume of internet searches or the number of social network posts)
  - sentiment as an emotional concept represented by a dicrete or continuous score variable (Groth and Muntermann, 2011; Chan and Chong, 2017)

Textual analysis and applications in Finance II

The development of automated processes for text mining of news and social media has boosted empirical research within this approach.

Most of the research is devoted to computing the investor sentiment (or attention) about a specific asset.

This is usually done using predefined dictionaries and available algorithms, such as VADER (Hutto and Gilbert, 2014), to obtain the polarity and/or degree of sentiment of the corpus of text analyzed.

However, the reliability of commonly available sentiment algorithms has been questioned in different studies and economic fields (Jussila et al., 2017; Colladon et al., 2020; Gloor et al., 2022).

Part 1: The semantic importance score

The semantic importance of words I

In Fronzetti Colladon (2018) a novel methodology has been introduced to evaluate the **semantic importance** of a word within a given text.

This approach is an alternative to measuring the favorability and complements the measure of frequency.

In fact, the semantic score is based on the relationships among the words appearing in the given text that are modeled through a network.

The basic idea behind the semantic importance score is that:

"The greater the frequency of a word within a discourse, and the more it is enriched with associations, the more it will be retained and possibly increase its potential to resonate with the intended audience", (Fronzetti Colladon, 2018).

The semantic importance of words II

The considered features of each keyword are:

- **prevalence**, which measures how frequently a keyword is used in the text
- **diversity**, which evaluates how heterogeneous and strong are the associations to a keyword.
- **connectivity**, which highlights how much that concept can bridge connections among other keywords in the text.

The semantic importance of words III

Prevalence is calculated as the frequency of a keyword in a given set of documents and time-frame and is of common use in textual analysis.

It measures how frequently a specific keyword is used in the text flow: the more a word appears, the more readers will remember and recognize it, which could ultimately influence their opinions and behaviors.

However, the importance of a keyword depends not only on its frequency of occurrence but also on its association with other keywords in the text.

To account for these associations, the SBS indicator includes the other two dimensions, stemming from the theory of *social network analysis* (SNA), which help evaluating how heterogeneous and strong are the links to a keyword and how much that keyword can bridge connections among other keywords in the discourse.

The semantic importance of words IV

The same principle, the same love of system, the same regard to the beauty of order, of art and contrivance, frequently serves to recommend those institutions which tend to promote the public welfare.

Adam Smith - The Theory of Moral Sentiments, 1759

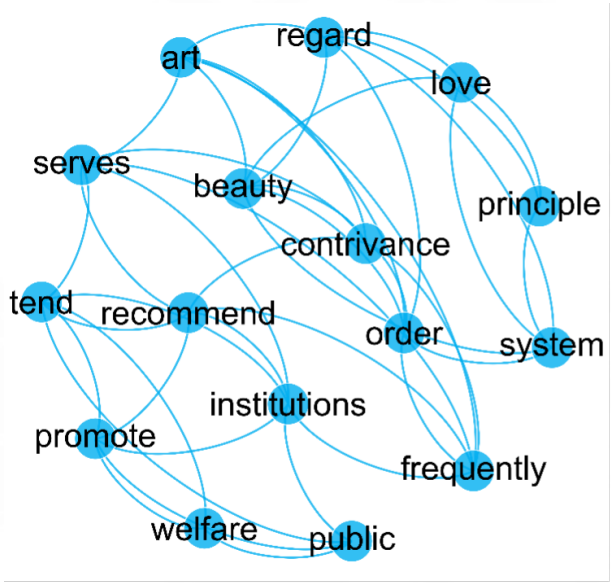


Figure: An example of output visualized using the following sentence by Adam Smith

For the sake of readability, in the picture nodes are labeled before stemming.

The semantic importance of words V

A network is formalized mathematically by a graph G, with n nodes (keywords clusters) and E edges, associated with a set of connection weights represented by a matrix $W = \{w_{ij}\}$.

Diversity

The *Diversity* of a keyword i measures how heterogeneous and strong are its links and is defined through the following formula:

$$DI(i) := \sum_{j=1, j\neq i}^{n} \log_{10} \frac{n-1}{g_j} \mathbb{I}_{\{w_{ij}>0,\}}, \quad \text{for } i = 1, 2, \ldots, n$$

where g_j is the degree (number of connections) of node j, $j = 1, 2, \ldots, n$.

$\mathbb{I}_{\{w_{ij}>0\}}$ is an indicator function that is equal to 1 when the edge connecting nodes i and j exists, i.e., when $w_{ij} > 0$, and is equal to 0 when this edge is missing.

The semantic importance of words VI

The last dimension of the SBS, represents how much each keyword acts as a bridge to connect other terms and keywords in the text.

Connectivity

The *Connectivity* of keyword i is given by:

$$CO(i) := \sum_{j<k} \frac{d_{jk}(i)}{d_{jk}}, \quad \text{for } \; i = 1, 2, \ldots, n,$$

where d_{jk} is the number of shortest network paths connecting nodes j and k and $d_{jk}(i)$ is the number of those paths that include node i $(j, k = 1, 2, \ldots, n)$.

The semantic importance of words VII

The three dimensions are standardized and summed up to finally define the importance index of each keyword.

The Semantic Brand Score

The *Semantic Brand Score* of each keyword i is calculated by summing the standardized scores of $PR(i), DI(i), CO(i)$:

$$SBS(i) := \frac{PR(i) - \overline{PR}}{std(PR)} + \frac{DI(i) - \overline{DI}}{std(DI)} + \frac{CO(i) - \overline{CO}}{std(CO)}, \text{ for } i = 1, 2, \ldots, n,$$

where $\overline{PR}, \overline{DI}, \overline{CO}$ and $std(PR), std(DI), std(CO)$ are, respectively, the average value and the standard deviation across all keywords of the three dimensions involved in the score in chosen texts (NB: and time-frame in a dynamic setting).

Note that other standardization choices, such as subtracting the median and dividing by the interquartile range, yield similar results (Fronzetti Colladon, 2018).

Computing the semantic importance score

The computational process is performed in several steps:

- identification of a possibly extensive list of keywords related to the topic of interest in order to collect the most of the available information in the news media/forums/social networks.
- grouping of keywords with similar meaning in clusters in order to reduce dimension, if needed.
- common text pre-processing routines – such as tokenization, removal of stop-words, and removal of word affixes, known as stemming.
- designing a network between keywords and evaluating their features.
- finally assigning a score to measure the semantic importance of each keyword.

Case study on environment al climate-related keywords

The aim of the numerical experiment is to assess the semantic importance of the discourse on *environmental and climate topics* extracted weekly from a group of major US newspapers and to investigate whether and to what extent it impacts financial markets.

We collect all articles that appeared in *New York Times, Los Angeles Times, USA Today, Financial Times, Wall Street Journal*, from January 2014 to November 2022. Texts are analyzed separately for generalist and financial journals on a weekly time frame.

1. We select 13 sub-themes and define corresponding keywords related to climate and environment (clusters).
2. We compute the semantic importance score for each of the 13 clusters with a weekly time frame obtaining the SBS time series.

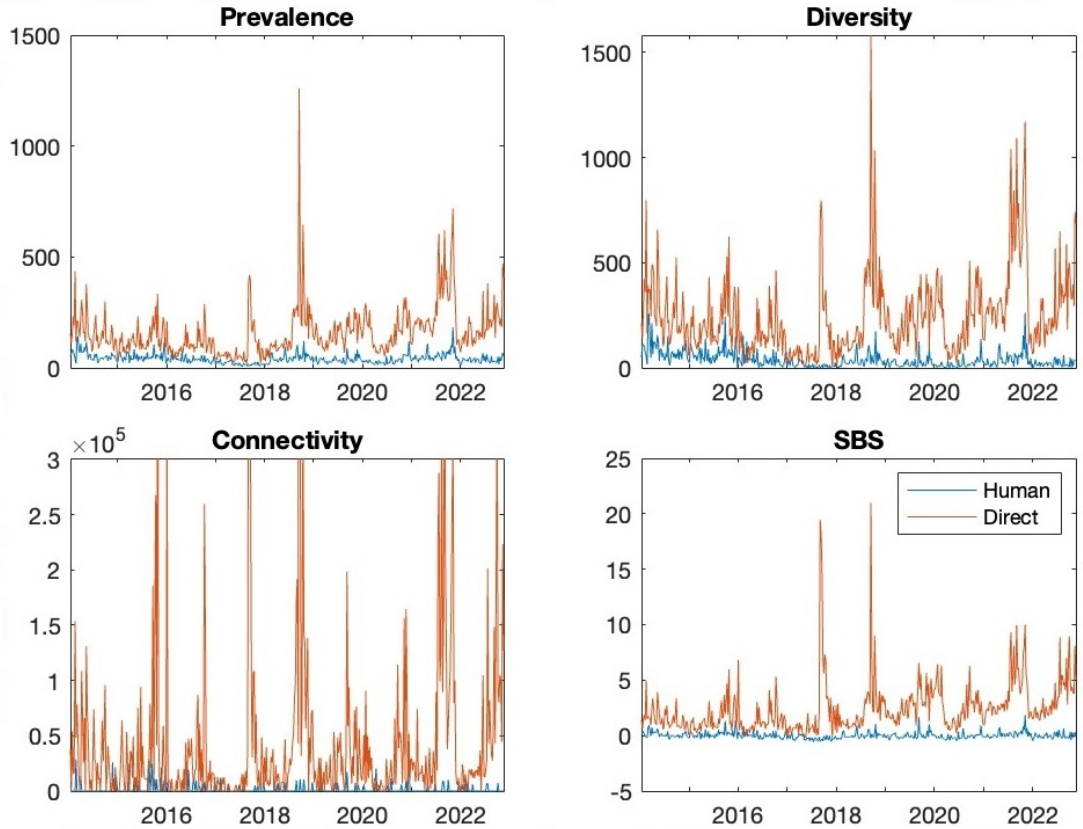
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Selected Clusters

Cluster	Description
Renewable Energy	Energy sources naturally replenished such as solar wind and hydro power alternatives to fossil fuels
Climate Action	Efforts to reduce greenhouse gas emissions and mitigate the effects of climate change.
Climate Human Effects	Effects of climate change caused directly by human actions
Direct Climate Change Effects	General effects of climate change
Waste Negative	Words related to negative waste management
Waste Positive	Words relate to positive waste management
Gas Emission	Gases released in the atmosphere and contributing to global warming
Energy Efficient	Words related to energy efficient
Sustainable Mobility	Refers to sustainable transportation
Net Zero Emissions	Balance between the amount of greenhouse gases emitted and the amount removed from the atmosphere
Environment&Education	Education topic declined to the concept of environment
Sustainability General	Generic words related to sustainability
Climate Resilience	Generic words related to the concept of climate resilience

Table: Selected cluster for environmental and climate-related words

Computing the SBS for selected clusters



Source: *Wall Street Journal*, *Financial Times*
(over 1500 articles per week in average).
Clusters: *Climate human effects*, and *Direct climate change effects*

Part 2: The construction of topic importance composite indexes.

Composite Indexes

Composite Indexes aim at quantifying a non directly measurable complex phenomenon (latent variable) by assuming that it can be broken down into K measurable components.

Given a multivariate dataset $X_1, ..., X_K$, of informative variables, a composite indicator (CI) is usually obtained by:

1. **Transformation** of the components to allow comparability:

$$T_j(X_j), \quad j = 1, \ldots, K$$

2. **Aggregation** with a suitable function

$$\Psi[T_1(X_1), \ldots, T_k(X_k);\ w_1, \ldots, w_k]$$

with $w_1, \ldots, w_k$ suitable weights representing the degree of relevance of the components.

Greco et al. (2019) present an interesting review on the methodology underneath Composite Indexes.

Composite Indexes: common weighting schemes

The most well-known aggregation of information is the **Mean** which averages single components (usually standardized).

A common alternative is to consider the first principal component $\mathbf{PC}_1$, obtained through Principal components analysis (PCA).

Drawback

Both standardization and distribution-related aggregating function may change the original data structure and the information summarized by each component relative to the target latent phenomenon.

For instance, the weighting scheme in PC_1 is obtained mathematically to maximize the "explanation" of the inner covariability of the system of standardized indicators which not necessarily associates to the relevance of the components with respect to the analyzed phenomenon.

Indeed, studies have shown that experts generally interpret weights as indicators of importance rather than trade-offs (Fusco, 2015; Greco et al., 2019).

Composite Indexes: Data-driven weighting scheme

To overcome issues with statistical models, new methodologies have been introduced such as the **Benefit of the Doubt (BoD)** method based on the Data Envelopment Analysis (DEA).

Assumptions

This approach builds on the idea that weights:

- are non-negative;
- may vary over time ($t = 1, 2, \ldots, T$) within an assigned range;
- need to respect an ordinal ranking defined through the relative efficiency (see below).

Endogenous Weighting of Components

Weights are obtained for each time-frame $t = 1, 2, \ldots, T$ as

$$(w^*_{1,t}, \ldots, w^*_{K,t}) := \underset{(w_{1,t},\ldots,w_{K,t})\in\mathcal{W}}{argmax} \quad \frac{\sum_{j=1}^{K} w_{j,t} X_{j,t}}{\max_h \sum_{j=1}^{K} w_{j,t} X_{h,t}}$$

with $\mathcal{W}$ representing the set of weights restrictions.

From BoD to BoD-Dir

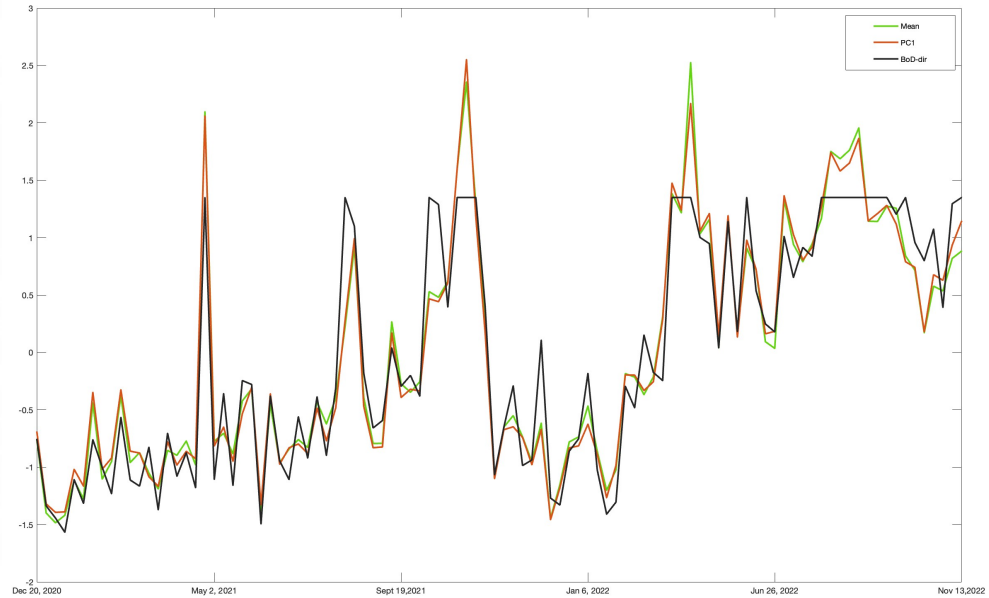
- BoD is extremely parsimonious about weighting assumptions and allows the data to decide on the weights over time.
- The BoD composite index are more sensitive to the original data structure with respect to statistical methods.

A penalty/reward can be added to the weights restrictions as suggested in Fusco (2015), where the preference structure is determined from the variability of component projected into principal components and the penalty criteria is based on directional distance functions

$\longrightarrow$ **Directional Benefit of Doubt** (BoD-dir).

Topic Composite Index construction

We compute the *Mean*, PC_1 and $BoD - dir$ across the 13 clusters with a weekly time-frame.



Time series of the considered Composite Indexes (rescaled for readability)
December 2020-November 2022

Summary Weights for Financial News

Variable	Freq.	Min	Max	Mean	Mean Overall	Variance	PC1
climateaction	59	0.003	1.00	0.215	0.027	0.039	0.095
climatehumaneffects	80	0.002	1.00	0.120	0.021	0.031	0.052
climateresilience	351	0.025	0.63	0.433	0.328	0.012	0.084
directclimateeffects	47	0.004	1.00	0.256	0.025	0.051	0.046
educationenv	346	0.006	1.00	0.303	0.225	0.008	0.076
energyefficient	319	0.056	1.00	0.188	0.129	0.012	0.087
greenhousegas	24	0.001	0.59	0.109	0.005	0.021	0.090
greenmob	170	0.003	1.00	0.166	0.060	0.046	0.076
netzeroem	59	0.002	1.00	0.123	0.015	0.039	0.102
renewableenergy	54	0.002	1.00	0.185	0.021	0.039	0.093
sustainabilityall	61	0.002	1.00	0.263	0.034	0.085	0.088
wasteneg	177	0.003	1.00	0.183	0.070	0.023	0.046
wastepos	102	0.017	1.00	0.178	0.039	0.019	0.065

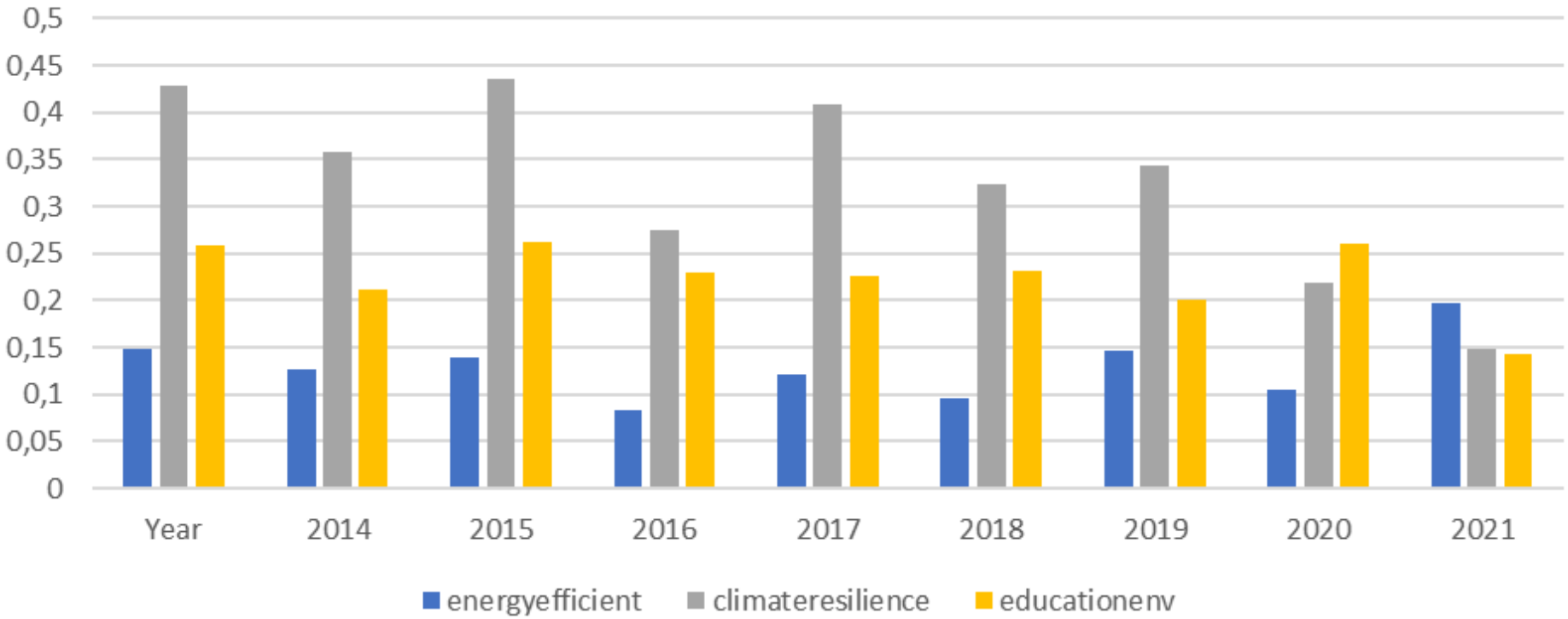
Table: Structure of BoD and PC_1 Index Weights based on Financial News (equal weights=0.076)

Summary Weights for General News

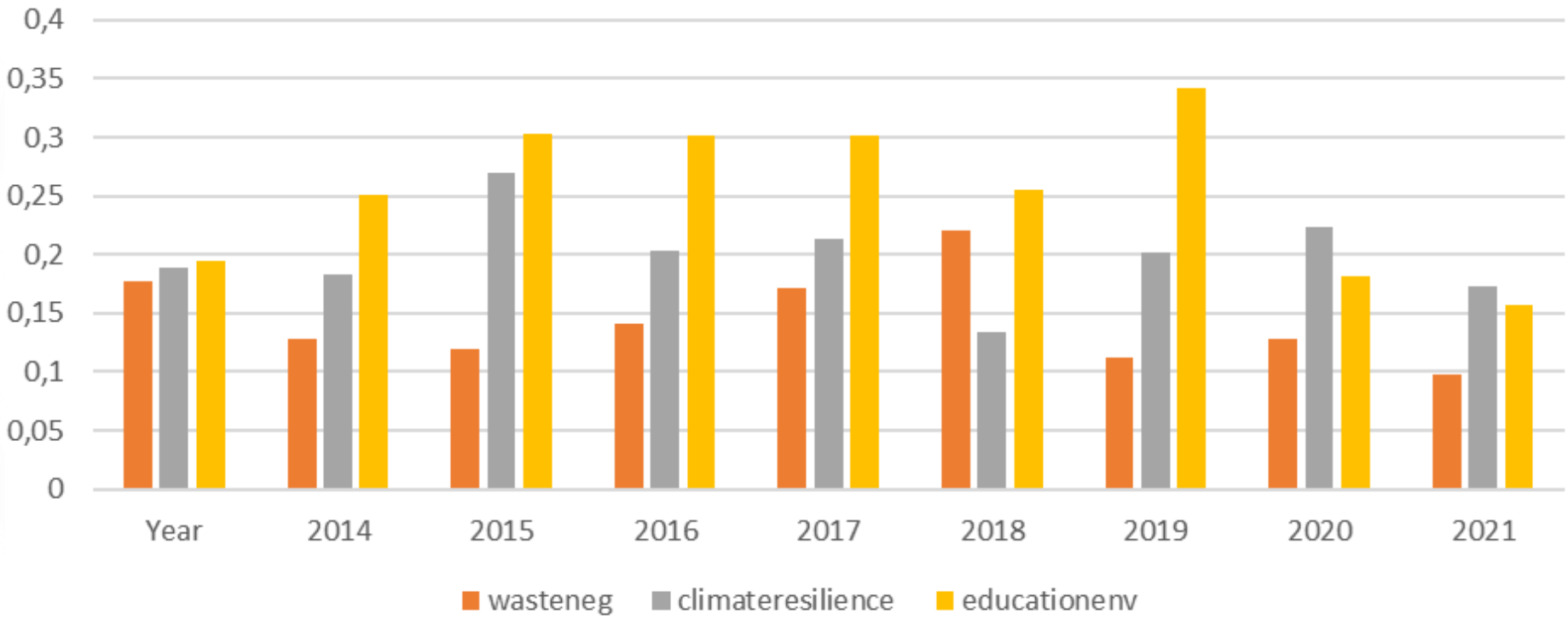
Variable	Freq	Min	Max	Mean	Mean Overall	Variance	PC1
climateaction	88	0.007	1.00	0.196	0.037	0.016	0.089
climatehumaneffects	131	0.005	1.00	0.162	0.045	0.012	0.062
climateresilience	340	0.000	1.00	0.272	0.199	0.015	0.101
directclimateeffects	171	0.041	1.00	0.190	0.070	0.006	0.021
educationenv	367	0.004	1.00	0.323	0.255	0.014	0.089
energyefficient	237	0.002	1.00	0.125	0.064	0.010	0.090
greenhousegas	140	0.002	1.00	0.124	0.037	0.012	0.079
greenmob	76	0.006	1.00	0.192	0.031	0.041	0.063
netzeroem	255	0.004	0.55	0.122	0.067	0.008	0.103
renewableenergy	56	0.002	1.00	0.144	0.017	0.020	0.091
sustainabilityall	66	0.001	1.00	0.162	0.023	0.019	0.098
wasteneg	316	0.002	0.68	0.213	0.145	0.013	0.054
wastepos	43	0.005	1.00	0.116	0.011	0.028	0.060

Table: Structure of BoD and PC_1 Index Weights based on General News (equal weights=0.076)

Comparison Weights Overtime (Financial)

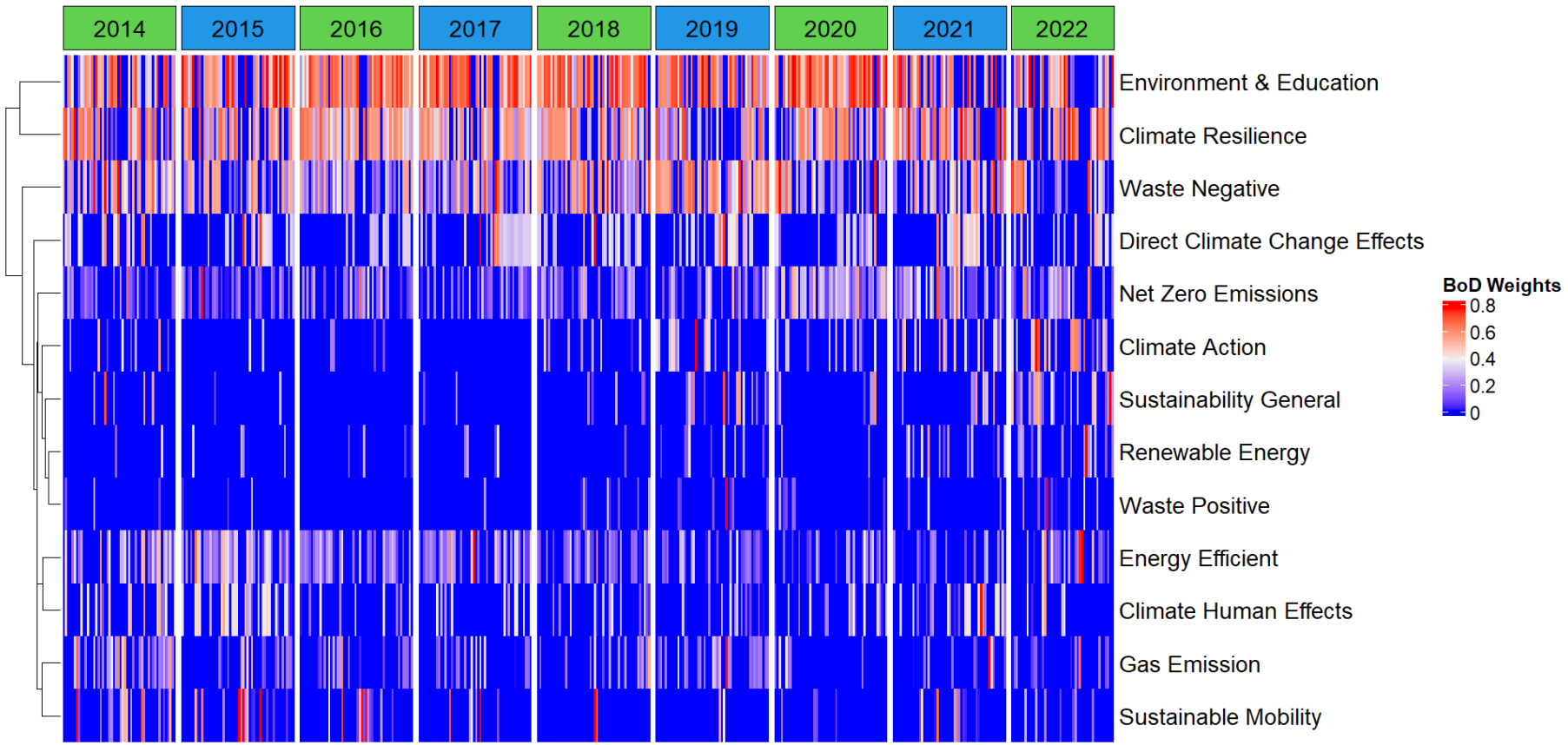


Comparison Weights Overtime (General)



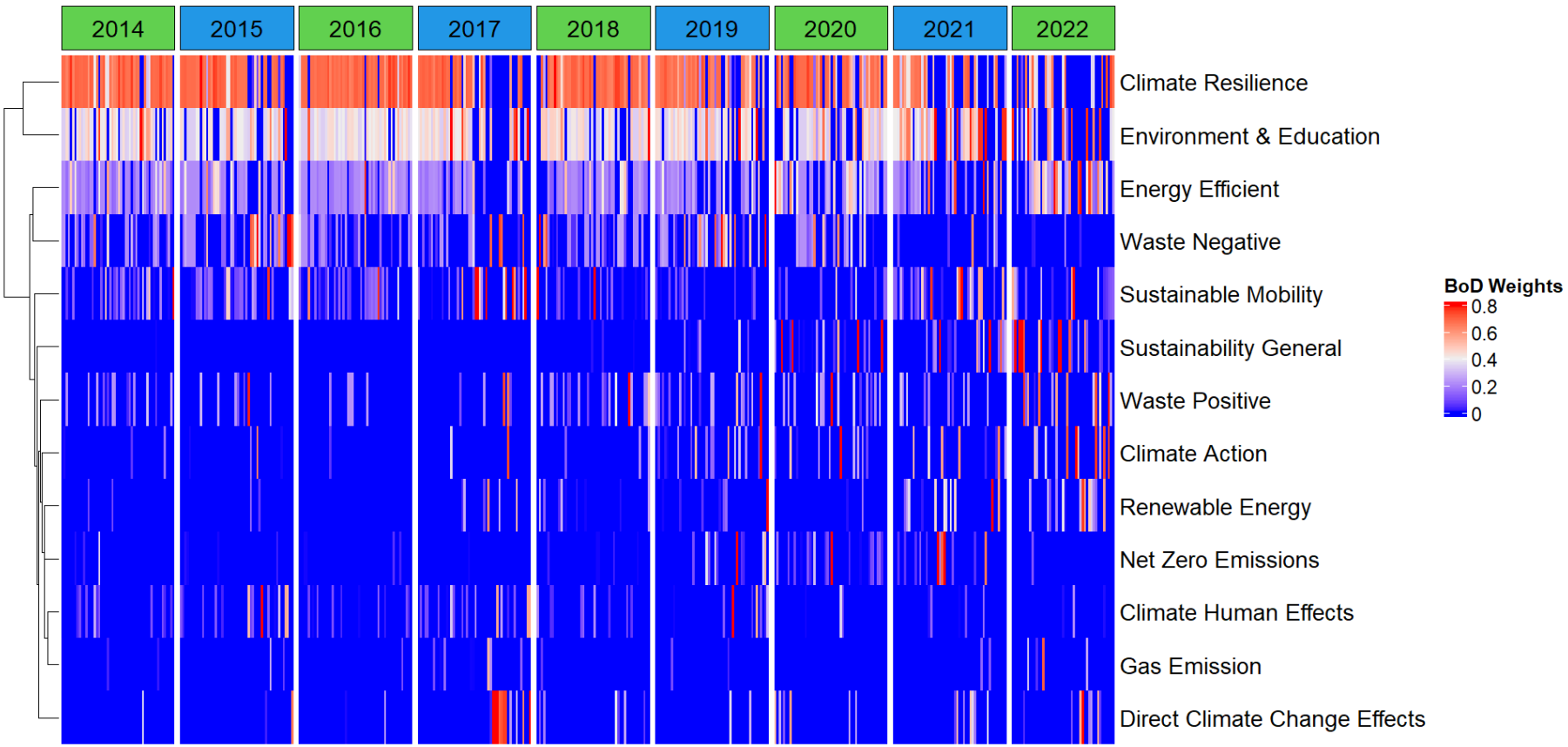
BoD Index weighting scheme over time

Generalist newspapers



BoD Index weighting scheme over time

Financial newspapers



Part 3: Assessing the impact of topic importance in financial markets.

Model specification

As a first experiment to verify whether the topic of interest affects financial markets we investigate if the dynamics of the returns of two market indexes, modeled by an EGARCH(1,1) process, is better described when the topic score is taken into account as an added exogenous factor.

$$R_t = a + \mathbf{b} X_t^r + \epsilon_t \ , \ t = 1, 2, \ldots, T;$$

where $R_t = \ln \frac{P_t}{P_{t-1}}$ is the logarithmic return, $\epsilon = \{\epsilon_t, t \geq 0\} = \sqrt{h_t}\eta_t$ is the error process with $\eta = \{\eta_t, t \geq 0\}$ Gaussian white noise and h_t is the so-called conditional variance:

$$\log h_t = \alpha_0 + \alpha_1 \eta_{t-1} + \beta_1 \log h_{t-1} + \lambda \left(|\eta_{t-1}| - \mathbb{E}\left[|\eta_{t-1}| \right] \right) + \boldsymbol{\gamma} X_t^h,$$

where $a, b, \alpha_0, \alpha_1, \beta_1, \lambda, \gamma$ are constant parameters.

$X^r = \{X_t^r, t \geq 0\}$ and $X^h = \{X_t^h, t \geq 0\}$ measure either the semantic importance indexes or their first differences.

Model specification

Table: Exogenous factors in model specification

X^r	X^h
Mean	Mean Diff
PC1	PC1 Diff
BoD-dir	BoD dir Diff

These models are compared with the benchmark model where no sentiment factor is accounted for ($b = \gamma = 0$).

Explanatory variables are represented by the CIs time series in the mean and by their first differences in the variance.

Note that CIs are computed considering either the generalist or the financial news on a weekly time-frame.

Assessing the impact on return and volatility dynamics

	Model 0 - No X	**Model 1** - Mean	**Model 2** - PC1	**Model 3** - BoD-dir
Panel a: General News				
a_0	0.0011	-0.0009*	0.0014	-0.0148***
b	-	0.0084***	0.0003	**0.0183*** **
α_0	-0.9877***	-0.9908***	-0.9808***	-0.9766***
α_1	-0.1641***	-0.1762***	-0.1743***	-0.1773***
β_1	0.8690***	0.8692***	0.8704***	0.8711***
λ	0.4027***	0.3903***	0.3907***	0.3949***
γ	-	-4.6206***	-0.1367***	**-4.6962*** **
$\mathcal{L}$	1141.415	1146.736	1146.381	**1147.945**
Panel b: Financial News				
a_0	0.0011	0.0005	0.0014	-0.0052***
b	-	0.0031	0.0002	**0.0081*** **
α_0	-0.9877***	-1.2062***	-1.1535***	-1.0807***
α_1	-0.1641***	-0.1673***	-0.1647***	-0.1473**
β_1	0.8690***	0.8412***	0.8478***	0.8577***
λ	0.4027***	0.4800***	0.4576***	0.4895***
γ	-	4.5077***	0.1334*	**4.8166*** **
$\mathcal{L}$	1141.415	1145.419	1143.739	**1149.170**

Table: S&P 500 parameter estimates

Assessing the impact on return and volatility dynamics

	Model 0 - No X	**Model 1** - Mean	**Model 2** - PC1	**Model 3** - BoD-dir
Panel a: General News				
a_0	-0.0021	0.0011	-0.0021	-0.0145***
b	-	-0.0120	-0.0004	**0.0150*** **
α_0	-0.3082**	-0.3387**	-0.3436**	-0.3184***
α_1	-0.1475***	-0.1375***	-0.1368***	-0.1329***
β_1	0.9493***	0.9445***	0.9437***	0.9484***
λ	0.3070***	0.3156***	0.3188***	0.2916***
γ	-	-2.3356	-0.0862	**-4.8704*** **
$\mathcal{L}$	862.547	864.0352	864.508	**866.579**
Panel b: Financial News				
a_0	-0.0021	-0.0113***	-0.0010	-0.0196
b	-	**0.0360*** **	0.0014*	0.0207
α_0	-0.3082**	-0.3303***	-0.3296***	-0.3382***
α_1	-0.1475***	-0.1451***	-0.1462***	-0.1393***
β_1	0.9493***	0.9468***	0.9469***	0.9450***
λ	0.3070***	0.3116***	0.3089***	0.3245***
γ	-	1.7921	0.0611	**2.4199***
$\mathcal{L}$	862.547	866.156	866.096	**866.158**

Table: S&P 500 Energy parameter estimates

Concluding remarks I

Where we are now:

1. **Methodology**

   We constructed several topic semantic importance indexes by suitably aggregating the semantic importance score for selected keywords/clusters.

2. **Case study in financial markets**

   We evaluated the relevance of environmental and climate issues (as a theme of interest in the news) on the dynamics of returns of two US stock market indexes.

Concluding remarks II

Future directions:

1. **Methodology**
   - We would like to give a comprehensive comparison of the proposed composite indexes with alternative indicator $\longrightarrow$ *timeframeissues*;
2. **Application in financial markets**
   - We plan to perform an out-of-sample exercise to investigate the relevance of environmental and climate issues (and possibly other topical themes in the news) in the forecasting of returns.
   - In addition, we would like to assess whether the importance of topics in the news can help to build hedging strategies (by applying the above model estimation to multiple assets or sectors) as in Engle et al. (2020); however, the index is available only until 2017 on a monthly basis and direct comparisons are not reliable.

Thanks for your attention!

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